Debanjan Das

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	8.03
SRI AUROBINDO VIDYAMANDIR, CHANDERNAGORE	Class XII (WBCHSE)	2019	87.40%
SRI AUROBINDO VIDYAMANDIR, CHANDERNAGORE	Class X (WBBSE)	2017	90.00%

PROJECTS

Adversarial Attacks on Autonomous Vehicles || M.Tech Research Work | Github

- Developed a robust Virtual Test Bed in CARLA to simulate and evaluate the efficiency of Physical Adversarial Attacks under diverse environmental conditions.
- Engineered custom adversarial patches specifically designed to exploit the architectural vulnerabilities of the **latest object detection** pipelines, including **YOLOv11, v12, and v26.**
- Synthesized patch-generation methodologies influenced by state-of-the-art research from ICCV 2021 and ICLR 2025, optimizing for high transferability and physical-world persistence.
- Created a **comprehensive model benchmarking API** in CARLA to quantify the **drop in Mean Average Precision (mAP)** across the YOLO family under adversarial attack.

Thermal Image Object Detection | Github

- Finetuned a modified **YOLOv8m** on a dataset of 1 million thermal images to detect multiple objects.
- Engineered **3-channel PCA temporal embeddings** to efficiently deal with thermal drift.
- Implemented spatial **Convex Hull filtering** to suppress false positive hallucinations aggressively.
- **Results**: Achieved a Global **mAP@50 of 0.59** and a Balanced Score of 0.498, resulting in a **14% improvement** in weighted mAP and a **23% improvement** in global mAP over the baseline score.

Noisy Image Restoration | *Runners Up,* *KLA x IITH AI Hackathon* | Github

- Developed an efficient **NAFNet**-based architecture for compound image restoration (deblurring and super-resolution).
- Formulated a physically-motivated **Heteroscedastic Data Consistency (DC) Loss** to better model the degradation pipeline.
- Bridged low-resolution noisy observations with reconstructed outputs via a heavy-tailed **Student's-t formulation**, which significantly boosted the model robustness to outliers, mixed noise distributions, and optimization variance collapse.
- Maximized deployment efficiency and inference speed by building the training pipeline with **PyTorch DDP + AMP** utilizing lightweight activation-free restoration blocks, and implementing TTA-based refinement strategies.
- **Results**: Secured a score of 0.88303(benchmark score- 0.89), leading to secure a **Runners-up position** within IIT Hyderabad.

POSITIONS OF RESPONSIBILITY

Placement Coordinator

- Co-designed the official placement brochure and led targeted outreach, contacting **15+** prospective corporate partners for the upcoming recruitment drive.
- Directed the end-to-end hiring cycle for a designated corporate partner during the **Japan Career Day** event, owning the operational process from pre-placement presentations to candidate assessment logistics.

Teaching Assistant

- **Foundations of Machine Learning(AI5000)**
- Managed a class of 150+ students; designed technical assignments and quizzes to assess conceptual understanding.
- **Artificial Intelligence in Emerging Technologies(AIET, Collaboration between IIT H and Accenture)**
- Taught Computer Vision to a cohort of 40+ industry professionals, bridging theoritical concepts with practical

applications.

SKILLS

Languages: C, C++, Python

Machine Learning & AI: PyTorch, Transformers, CLIP, Hugging Face

Frameworks & Libraries: NumPy, Pandas, Matplotlib, OpenCV

Tools & Infrastructure: Git, TensorBoard

Databases: MySQL

RELEVANT COURSES

Artificial Intelligence and Machine Learning: Foundations of Machine Learning, Matrix Theory, Deep Learning, Computer Vision, Representation Learning.

Computer Science : Data structures and Algorithms.